Mahdi Rastegar

Toronto, ON | +1-647-782-3513 | m.rastegardoost@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

PhD-qualified AI solutions engineer and applied ML researcher with 8+ years of experience developing data-driven engineering systems from research concept through prototype, validation, and deployment readiness. Builds practical machine learning, generative AI, and agentic workflows for complex engineering problems, combining Python software, production-quality APIs, automated data pipelines, robotics, embedded controls, and physics-based simulation. At the Acceleration Consortium, led an AI-enabled self-driving laboratory platform that increased throughput by 400%, reduced cycle time by 75%, and processed 1.7M data points per 24-hour campaign. Brings strong mechanical engineering domain knowledge, technology evaluation, scientific communication, and cross-functional R&D leadership.

TECHNICAL SKILLS

AI, ML & GenAI: Machine learning principles, supervised learning, Random Forest, neural networks, feature engineering, active learning, Bayesian optimization, uncertainty quantification, physics-informed modeling, time-series analysis, GPT-4/Claude API integration, prompt engineering, agentic workflow orchestration
Frameworks & Programming: Python, PyTorch, scikit-learn, NumPy, Pandas, SciPy, C++, JavaScript, SQL, object-oriented programming, automated testing
Production AI Systems: RESTful APIs, Flask, SocketIO, WebSocket services, real-time dashboards, reproducible data pipelines, model validation, automated reporting, experiment tracking, anomaly detection
Engineering Software: ANSYS Fluent, SolidWorks, Fusion 360, CFD, simulation-to-test correlation, design of experiments, statistical validation, uncertainty-aware decision support
Automation & Integration: Multi-agent hardware/software orchestration, OpenTrons OT-2, ESP32/ESP8266, sensor networks, CNC systems, instrumentation, real-time control, fault handling
Development Environment: Git/GitHub, Linux terminal, Bash scripting, Jupyter, debugging, logging, documentation, collaborative software development
Engineering Leadership: Technology scouting and evaluation, requirements definition, concept development, prototyping, verification and validation, manufacturability, technical risk, project coordination, stakeholder communication

PROFESSIONAL EXPERIENCE

Research Associate & AI Solutions Project Lead | Acceleration Consortium, University of Toronto **Jan 2025 - Present**

- Architected an AI-enabled self-driving laboratory platform integrating active learning, Bayesian optimization, uncertainty quantification, robotic execution, embedded controls, and real-time data services; increased parallel testing capacity from 2 to 8 and reduced experiment time by 75%
- Led concept-to-validation development of a production-oriented engineering platform spanning mechanical design, electrical/control integration, software architecture, experimental design, and operational qualification
- Built Python-based orchestration, REST API, and WebSocket monitoring services coordinating an OpenTrons OT-2 robot, ESP8266 sensor network, pumps, thermal systems, and custom test hardware, reducing manual intervention by 90%
- Developed reproducible data pipelines and automated quality checks processing 1.7M data points per 24-hour campaign; achieved 99.5% data accuracy and enabled rapid anomaly detection and technical decisions
- Applied ML-assisted experimental design and uncertainty-aware optimization to prioritize experiments under resource constraints, achieving a 95% autonomous operation success rate with zero cross-contamination
- Coupled ANSYS Fluent simulation with experimental calibration to optimize an 8-channel apparatus, achieving 98% flow uniformity and 35% pressure-drop reduction while maintaining USP-aligned performance
- Coordinated a multidisciplinary team across mechanical, electrical, software, and research scopes; managed milestones, technical risks, documentation, progress reporting, and transition of validated concepts to internal users
- Evaluated emerging AI, automation, instrumentation, and engineering technologies through scientific analysis, application fit, performance trade-offs, and prototype testing

Postdoctoral Fellow | Acceleration Consortium, University of Toronto **Apr 2024 - Dec 2024**

- Developed AI-guided autonomous experimentation workflows combining active learning, Bayesian optimization, and robotics to improve engineering test planning and resource allocation
- Built Python-based data analysis and monitoring tools for high-throughput experiments, translating experimental results into engineering recommendations and validated design changes
- Collaborated with researchers, engineers, and external partners to evaluate emerging technologies and mature proof-of-concept concepts toward deployable R&D capabilities

Research Appointee | KITE Research Institute, University Health Network **Jan 2024 - Feb 2025**

- Designed data-driven experimental programs for advanced porous composites, linking processing, structure, thermal, mechanical, transport, and sensing results to engineering down-selection decisions
- Built and analyzed multi-modal characterization datasets and translated results into reproducible engineering guidelines for scale-up,

reliability, and application-oriented prototyping
- Developed aerogel systems with service stability up to 575°C, 91.8% capacitance retention after 5,000 cycles, and improved sorption and sensing performance through structured optimization

Research Assistant | University of Toronto **Jan 2021 - Jan 2024**
- Developed and validated multifunctional sensing and energy systems through experimental design, physical modeling, signal analysis, statistical validation, and iterative engineering optimization
- Created lightweight materials and sensing platforms for harsh-environment monitoring, achieving 350,000+ cycle durability and quantified performance across broad thermal, mechanical, and acoustic conditions
- Published and presented research that converted complex scientific results into design rules, technical reports, and actionable recommendations for interdisciplinary collaborators

Mechanical Engineer | TECHNE KIROW GmbH, Germany **Aug 2019 - Sep 2020**
- Supported design, installation, commissioning, inspection, and maintenance of industrial thermal, pressure, hydraulic, and vehicle systems in standards-driven manufacturing environments
- Performed root-cause investigations and coordinated technicians, suppliers, and operations stakeholders to close technical issues under safety, quality, and schedule constraints
- Modified chassis, suspension, and hydraulic systems for steel-plant slag pot carriers, strengthening engineering judgment across mechanical design, manufacturability, and field reliability

SELECTED AI SOLUTIONS PROJECTS

Agentic AI and Autonomous Engineering Workflow
- Integrated GPT-4/Claude-based hypothesis generation and scientific knowledge workflows with active learning, Bayesian optimization, and tool-enabled experimental execution for autonomous R&D
- Designed a shared orchestration layer for LLM planning, robotic manipulation, sensor acquisition, and engineering validation, with human review points and physics-based quality gates
- Developed AI-assisted code-generation and analysis workflows with automated tests and domain validation to accelerate research software development while preserving technical correctness

Machine Learning for Digital Viscometry and Process Optimization
- Engineered feature pipelines from torque, displacement, flow, and temperature time series; trained Random Forest and regression models for viscosity inference with CV below 3%
- Applied Bayesian path optimization and intelligent sequencing to reduce an 18-sample measurement campaign from 6 hours to 2.5 hours, a 58% reduction
- Deployed Flask, SocketIO, and Plotly monitoring services for live status, anomaly tracking, remote diagnostics, and automated reporting

EDUCATION

Doctor of Philosophy (PhD), Mechanical & Industrial Engineering | University of Toronto **Jan 2021 - Jan 2024**
Dissertation: Multifunctional porous fluoropolymer films for triboelectric nanogenerator applications. Focus: engineering design, automation-enabled experimentation, statistical analysis, physical modeling, and advanced testing.

Master of Science (MSc), Mechanical Engineering | University of Tehran **Mar 2016 - Sep 2018**
Thesis: Inverse heat-transfer analysis for structural health monitoring and defect characterization. Research included thermal systems, simulation, optimization, solar-electric vehicle design, and battery thermal management.

Bachelor of Science (BSc), Mechanical Engineering | University of Tehran **Mar 2012 - Sep 2016**

RESEARCH IMPACT

- **15+ peer-reviewed publications**, 10+ international conference presentations, 300+ citations, and h-index 9 in materials, sensing, automation, and engineering systems
- Sanofi iDEA-TECH Awards-supported autonomous experimentation program; reviewer for materials and engineering journals
- Experienced communicator across research, engineering, software, manufacturing, supplier, and academic stakeholders